SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25

CO1 Statement: Apply advanced methodologies and agile project management techniques to manage software projects effectively

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Date:10/10/2026 Duration: 60 minutes

Code of Conduct

I **MANISH KUMAR L/1924260185** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: manish kumar

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an Agile Sprint Planning and User Story Workshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects.

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1. Identify the Project Vision and Stakeholders

- Define the project's vision, objectives, and identify all key stakeholders involved.

2. Create Epics and User Stories

- Break down the requirements into Epics and formulate well-structured User Stories using the format: As a <user>, I want <feature>, so that <benefit>.

3. Define Acceptance Criteria

- Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

4. Prioritize the Product Backlog

- Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

5. Plan the Sprint Backlog

- Select high-priority user stories for the first sprint and divide them into actionable development tasks.

6. Estimate Effort Using Story Points

- Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

7. Present the Sprint Goal and Expected Deliverables

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Question

Intelligent Digital Twin Monitoring Platform for Industrial Assets

Industry Context

Industrial organizations need continuous visibility into complex assets such as turbines, pumps, compressors, and production systems.

Problem Statement

Design and develop a digital-twin platform that maintains virtual representations of physical assets and synchronizes operational, maintenance, sensor, and performance information.

Objectives

- Create digital asset models.
- Synchronize sensor data.
- Monitor equipment status.
- Maintain maintenance history.
- Visualize asset performance.
- Identify abnormal conditions.

Expected Outcomes

- Improved asset visibility.
- Better maintenance planning.
- Faster fault identification.
- Reduced downtime.
- Improved operational decision-making.

Answer

1. Identify the Project Vision and Stakeholders

Project Vision: Provide industrial organizations with a unified, reliable digital-twin platform that represents each physical asset, synchronizes sensor and operational information, preserves maintenance history, and helps personnel understand performance and identify abnormal conditions early.

Objectives:

- Create digital representations for turbines, pumps, compressors, and production systems.
- Synchronize sensor, operational, maintenance, and performance information with the relevant digital twin.
- Monitor equipment status and visualize performance trends.
- Maintain accurate, traceable maintenance history.
- Identify abnormal conditions and present alerts to responsible personnel.
- Support better maintenance planning, faster fault identification, reduced downtime, and informed decisions.

Key stakeholders and responsibilities:

- Plant/Operations Managers – review asset availability, performance dashboards, and KPIs.
- Maintenance Engineers and Technicians – investigate faults, review history, and record inspections and repairs.
- Reliability Engineers – analyze trends and recurring abnormal conditions.
- Control Room Operators – monitor live equipment status and respond to alerts.
- IoT/Data Engineers – connect sensors, validate data flows, and maintain synchronization.
- Software Developers and Testers – implement modules, integrations, dashboards, and tests.
- System Administrators/Security Personnel – manage accounts, permissions, configurations, backups, and security.
- Management – use consolidated reports to plan resources and maintenance priorities.

2. Create Epics and User Stories

Epics group the platform's main capabilities. Stories follow the required Agile format.

Epic	ID	User Story
E1 Asset Management	US1	As a plant administrator, I want to register an industrial asset and define its attributes, so that the platform maintains a digital representation of each asset.
E2 Sensor Synchronization	US2	As an IoT engineer, I want to map incoming sensor streams to the correct asset, so that readings update the relevant digital twin.
E3 Monitoring	US3	As an operations manager, I want to view current asset status and performance trends, so that I can understand equipment condition quickly.
E4 Maintenance History	US4	As a maintenance technician, I want to record and retrieve inspections and repairs, so that maintenance actions are traceable.
E5 Abnormal Detection	US5	As a reliability engineer, I want the platform to flag readings that breach configured thresholds, so that unusual

		conditions can be investigated early.
E6 Alerts & Reports	US6	As a control room operator, I want prioritized alerts and asset reports, so that I can respond to important conditions and communicate status.

3. Define Acceptance Criteria

US1 – Register Asset

- Given an authorized administrator is signed in, when valid asset details and a unique ID are submitted, then the system saves the asset and displays its digital-twin profile. Missing fields or duplicate IDs produce a validation message.

US2 – Map Sensor Data

- Given a registered asset and supported sensor source, when an authorized user maps the sensor, then valid new readings are linked to the correct twin and timestamped. Invalid or unmapped readings are logged.

US3 – Monitor Performance

- Given sensor readings are available, when an authorized user opens an asset dashboard, then current status, latest readings, timestamps, and available trends are displayed. Stale or unavailable data is clearly indicated.

US4 – Maintenance History

- Given an authorized technician opens an asset, when a complete maintenance record is submitted, then it is saved with date, activity, and responsible person and appears in history.

US5 – Detect Abnormality

- Given a sensor threshold is configured, when a valid reading crosses it, then an abnormal event linked to the asset is created with value and time. Readings within range do not trigger threshold alerts.

US6 – Alerts and Reports

- Given events or asset records exist, when an authorized user filters alerts or requests a report, then matching results with asset ID and timestamps appear. Unauthorized users cannot access restricted information.

4. Prioritize the Product Backlog

Priority is based on business value, dependencies, and the need to establish an end-to-end foundation. P1 is highest.

Rank	ID	Backlog Item	Priority	Reason
1	US1	Register asset and create twin	P1	Foundation for all asset-linked features.
2	US2	Map and synchronize sensor data	P1	Supplies current data to the twin.
3	US3	Display status and trends	P1	Delivers immediate asset visibility.
4	US4	Record maintenance history	P2	Adds traceability and supports planning.
5	US5	Detect threshold abnormalities	P2	Enables earlier investigation; depends on sensor data.
6	US6	Alerts and reports	P2	Improves response after core data flows work.

5. Plan the Sprint Backlog

Assumed Sprint 1 duration: two weeks. The sprint focuses on a demonstrable foundation: register assets, associate sensor readings, and view basic status. Confirm capacity during sprint planning.

Story	Task ID	Actionable Development Task	Owner Role
US1	T1.1	Define asset fields, validation rules, and unique identifier.	Developer
US1	T1.2	Implement asset create/view functions and twin profile.	Developer
US1	T1.3	Test valid input, missing fields, and duplicate IDs.	Tester
US2	T2.1	Define sensor-to-asset mapping and reading data structure.	IoT Engineer
US2	T2.2	Implement ingestion validation, timestamping, and mapping.	Developer/IoT
US2	T2.3	Test valid, stale, invalid, and unmapped readings.	Tester
US3	T3.1	Create dashboard showing latest readings and status.	Developer
US3	T3.2	Add timestamps and stale/unavailable-data indicators.	Developer
US3	T3.3	Run end-to-end test from registration to dashboard.	Team

6. Estimate Effort Using Story Points

Technique: Planning Poker using a modified Fibonacci sequence (1, 2, 3, 5, 8, 13). Points represent relative complexity and uncertainty, not fixed hours. The team should agree final estimates during planning.

ID	User Story	Points	Justification
US1	Register asset / digital twin	3	Form, validation, and a small data model.
US2	Synchronize sensor data	5	Mapping, ingestion validation, timestamps, and integration tests.
US3	Basic status dashboard	5	Data retrieval, display, trends, and stale-data handling.
US4	Maintenance history	5	Record creation, retrieval, validation, and access rules.
US5	Threshold abnormality detection	5	Threshold configuration, event generation, and testing.
US6	Alerts and reports	8	Filtering, prioritization, report formatting, and authorization.

Sprint 1 selected stories: US1 (3) + US2 (5) + US3 (5) = 13 story points. This is provisional and should be adjusted to the team's velocity and available capacity.

7. Present the Sprint Goal and Expected Deliverables

Sprint Goal: Deliver the first working foundation of the platform so an authorized user can register an industrial asset, associate sensor readings with it, and view its latest status and basic performance information.

Expected Sprint 1 deliverables:

- Asset registration and digital-twin profile with required-field and duplicate-ID validation.
- Sensor-to-asset mapping and data-ingestion flow with timestamps and validation.
- Basic dashboard showing latest readings, status, and data freshness.
- Test cases/results for registration, mapping, invalid data, and end-to-end display.
- Setup/configuration notes and a sprint review demonstration.

Definition of Done: Selected stories are implemented, reviewed, tested against acceptance criteria, integrated into the working build, documented where needed, and demonstrated to the product owner or stakeholder. Unfinished work and limitations are recorded in the backlog.

Expected project-level outcomes include improved asset visibility, better maintenance planning, faster fault identification, reduced downtime, and improved operational decision-making. Sprint 1 establishes the foundation and does not deliver every advanced monitoring capability.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25